

Spectrum and Frequency of BRAF Mutations in Skin Melanomas in the Dalmatian Region of Croatia.

PMID: 38946192 | Indexed in PubMed

Mutation of the BRAF oncogene is one of the most common mutations detected in human neoplasia, occurring in 40-60% of all cutaneous melanoma. BRAF is a serine/threonine protein kinase which is an essential part of the mitogen-activated protein kinase (MAPK) pathway. It is physiologically activated by RAS, but in mutated form, due to molecular conformational change, BRAF becomes constitutively active with subsequent persistent activation of downstream cytoplasmic and nuclear proteins (MEK, ERK, ETS), which finally leads to gene expression that promotes cell growth and survival. Inhibition of the altered MAPK pathway by BRAF inhibitors and combined BRAF/MEK inhibitors in BRAF mutated melanoma has become a standard therapeutic approach (1,2). We recently reported the frequency and clinicopathological features of BRAF V600E mutated melanomas in the Dalmatian region of Croatia. This report included 80 cutaneous melanomas with BRAF analyses performed at our institution until the second half of 2017, using a kit which detected only BRAF V600E mutation (3). From the second half of 2017, we started using a kit which detects several types of BRAF mutations along with NRAS mutation. The aim of this report was to determine the spectrum and frequency of different BRAF mutations in a group of skin melanomas in the Dalmatian region of Croatia and to comment on the relationship between type of BRAF mutation and therapeutic response to MAPK pathway inhibition. The analysis included 179 patients with stage 3 and stage 4 cutaneous melanoma with known BRAF/NRAS mutational status. The paraffin blocks were forwarded from four Dalmatian hospitals (Split: 139 cases, Zadar: 17 cases, Šibenik: 13 cases, Dubrovnik: 10 cases). BRAF/NRAS mutation analysis was performed at the Institute of Pathology, Clinical Hospital Center Split, Croatia, in the period from the second half of 2017 to the end of 2022. For DNA extraction analysis, hematoxylin and eosin stained slides from each submitted sample were reviewed by a pathologist, and tumor tissue was identified for analysis. For all tissue specimens, DNA was extracted from sections (10 mm thick) using the cobas® DNA Sample Preparation Kit (Roche Molecular Diagnostics), following the manufacturer's protocol. The amount of genomic DNA was quantified using the Qubit® 2.0 Fluorometer (Life Technologies) and adjusted to a fixed concentration to be added to the amplification/detection mixture. For mutation analysis, the target DNA was amplified and detected on the cobas z 480 analyzer using the amplification and detection reagents provided in the Roche BRAF/NRAS mutation test (LSR) kit, according to the manufacturer's protocol. The test results were reported as follows: BRAF exon 11 mutation detected, BRAF V600E/E2/D mutation detected, BRAF V600K mutation detected, BRAF V600R mutation detected, BRAF K601E mutation detected, NRAS (G12X, G13X, A18T, Q61X, other NRAS Ex3/4) mutation detected, mutation not detected, or invalid result (no result was obtained on the cobas test). BRAF mutation was observed in 87 patients (48.6%), NRAS mutation was found in 27 patients (15.1%), while 65 patients (36.3%) were without BRAF/NRAS mutation (Table 1). In the group of BRAF mutated melanomas, 61 cases (70.1%) had V600E/E2/D mutation, 20 cases (23%) had V600K mutation, 3 cases (3.4%) had exon 11 mutation, 2 cases (2.3%) had V600R mutation, and 1 case (1.2%) had K601E mutation (Table 2). The observed frequency of BRAF mutated melanomas in this study was similar to the frequency reported in our previous study (48.6% and 47.5%, respectively) (3). The vast majority were BRAF V600 mutations, while BRAF non-V600 mutations were rare (95.4% and 4.6%, respectively). In the group of BRAF V600 mutations, V600E/E2/D mutations predominated, followed by V600K mutations, while V600R mutations were rare. Greaves et al. reported similar frequency of BRAF V600 mutations in a group of 499 BRAF-mutated cutaneous melanomas, with V600E/E2/D mutations observed in 77.2% cases, followed by V600K mutations observed in 17.2% cases, and V600R mutations observed in 2.6% cases (4). BRAF non-V600 mutations (exon 11 and K601E mutations) were rarely observed in this study, confirming the findings of other authors (4,5). A three-class system of BRAF mutations was recently proposed that takes into account the differences in their kinase activity, with

class I containing mutants with high kinase activity and high response rate to BRAF and BRAF/MEK inhibitors. Class II BRAF mutations have lower kinase activity than class I mutants, but higher than wild-type BRAF, showing resistance to BRAF inhibitor monotherapy and sensitivity to MEK and BRAF/MEK inhibitors. Finally, class III BRAF mutations are characterized by low kinase activity and low response rate to targeted therapy (6). BRAF V600 mutations belong to class I mutations, which means that the large majority of BRAF-positive melanomas in this study (95.4%) were sensitive to targeted therapy. However, the sensitivity to targeted therapy is different among different class I BRAF mutations. While large randomized controlled trials on combined BRAF/MEK inhibition showed good overall response (63-68%) and improvement of progression-free survival (PFS) and overall survival (OS) for the melanomas with most prevalent V600E and V600K mutations, Menzer et al. showed lower response rate to MAPK pathway inhibition (45%) in the group of metastatic melanomas with BRAF V600 mutations other than V600E/K. The overall response rate to MAPK pathway inhibition in the same group of melanomas with BRAF non-V600 mutations (class II and III mutations) was only 18% (7). In our group of BRAF mutated skin metastatic melanomas, we found only 6 cases (6.9%) with expected lower response rate to MAPK pathway inhibition: 2 cases with V600R mutation (class I non-V600E/K mutation), 1 case with K601E mutation (class II mutation), and 3 cases with exon 11 mutation (class II and III mutations).

Combination Atezolizumab, Cobimetinib, and Vemurafenib as a Treatment Option in BRAF V600 Mutation-Positive Melanoma: Patient Selection and Perspectives.

PMID: 39099762 | Indexed in PubMed

The treatment landscape for advanced and metastatic melanoma has drastically changed in recent years, with the advent of novel therapeutic options such as immune checkpoint inhibitors and targeted therapies offering remarkable efficacy and significantly improved patient outcomes compared to traditional approaches. Approximately 50% of melanomas harbor activating BRAF mutations, with over 90% resulting in BRAF V600E. Tumors treated with BRAF inhibitor monotherapy have a high rate of developing resistance within six months. Combination therapy with MEK inhibitors helped to mitigate this treatment resistance and led to improved outcomes. Due to the up-regulation of PD-1/PD-L1 receptors in tumors treated with BRAF/MEK inhibitor therapy, further studies included a third combination agent, anti-PD-1/PD-L1 inhibitors. This triple combination therapy may have superior efficacy and a manageable safety profile when compared with single or double agent therapy regimens.

Mutation-Specific CRISPR Targeting with SaCas9 and AsCas12a Restores Therapeutic Sensitivity in Treatment-Resistant Melanoma.

PMID: 39387253 | Indexed in PubMed

Background: Melanoma remains one of the most challenging cancers to treat effectively with drug resistant remaining a constant concern, primarily with activating BRAF mutations. Mutations in the BRAF gene appear in approximately 50% of patients, 90% of which are V600E. Two frontline BRAF inhibitors (BRAFi), vemurafenib and dabrafenib, are frequently used to treat unresectable or metastatic BRAF V600E melanoma. Initial response rates are high, but soon thereafter, 70-80% of patients develop resistance to treatment within a year. A major mechanism of resistance is the generation of a secondary Q61K mutation in the NRAS gene. Methods: We have developed an approach in which a CRISPR-Cas

complex can be designed to distinguish between mutant genes enabling resistance to standard care in tumor cells and normal genomes of healthy cells. For the first time, we demonstrated the utility of two CRISPR-directed mutation-specific editing approaches to restore BRAFi sensitivity in BRAFV600E/NRASQ61K resistant A375 cells. Results: We utilize an AsCas12a protospacer adjacent motif site created by the NRAS Q61K mutation and the Q61K mutation in the critical seed region of an SaCas9 sgRNA for Q61K-selective targeting. We show here that both approaches allow for effective NRAS targeting of only mutated-Q61K and after CRISPR-directed Q61K-targeting, previously resistant A375 cells are re-sensitized to BRAFi treatment. Conclusion: Our data support the feasibility of the development of CRISPR-Cas therapeutic approaches to the treatment of melanoma. Successful therapeutic CRISPR-directed gene editing would enable both specific and efficient editing of a mutation-specific targeting approach eliminate concern for on- and off-target damage to the genomes of healthy cells.

The case for adjuvant BRAF-targeted therapy versus adjuvant anti-PD-1 therapy for patients with resected, high-risk melanoma.

PMID: 37345669 | Indexed in PubMed

The development of highly effective BRAF-targeted therapy and immune checkpoint inhibition for patients with advanced metastatic melanoma has transformed the treatment of this disease. More recently, these advances have moved into the resected, high-risk stage II and III settings. For patients with resected, BRAF-mutant stage III melanoma, there are no head-to-head data to support the use of BRAF-targeted therapy (specifically the combination of dabrafenib and trametinib) with either single-agent nivolumab or pembrolizumab. Because the relapse-free and distant metastasis-free survivals are similar in a cross-trial comparison, it is not clear what the best option for these patients is. In this piece, the authors argue on behalf of and against both approaches. PLAIN LANGUAGE SUMMARY: Two types of therapy exist for patients diagnosed with melanoma who have completed surgery and remain at high risk for tumor recurrence: (1) drugs that target the BRAF mutation (found in ~50% of patients) and (2) immunotherapy. There are no data showing that either approach is better than the other, so the choice of which therapy is best for an individual patient can be challenging. In this article, we make arguments for and against each option.

Novel approaches in melanoma prevention and therapy.

PMID: 24114495 | Indexed in PubMed

The incidence of cutaneous melanoma has risen at a rate significantly higher than that for other malignancies. This increase persists despite efforts to educate the public about the dangers of excess exposure to UV radiation from both the sun and tanning beds. Melanoma affects a relatively younger population and is notorious for its propensity to metastasize and for its poor response to current therapeutic regimens. These factors make prevention an integral component to the goal of decreasing melanoma-related mortality. Transformation of melanocytes into malignant melanoma involves the interplay between genetic factors, UV exposure, and the tumor microenvironment. The roles of UV radiation in the etiology of melanoma are mediated by both direct damage of DNA through formation of photoproducts and production of reactive oxygen species (ROS). Many of the promising antioxidant agents under development for the prevention of melanoma are derived from foodstuffs. B-Raf is a

member of the Raf kinase family of serine/threonine-specific protein kinases that plays a role in regulating the MAP kinase/ERKs signaling pathway. About 50 % of melanomas harbor activating BRAF mutations. BRAF mutations are found in 59 % of the melanomas arising in skin with intermittent sun exposure, such as trunk and arms, as compared with only 23 % of the acral melanomas, 11 % of mucosal melanomas, and 0 % of uveal melanomas. Two new agents, ipilimumab and vemurafenib, have been shown to improve outcome of advanced melanoma as presented at the plenary session of the 2011 annual meeting of the American Society of Clinical Oncology. Vemurafenib is the first personalized compound which demonstrated an improvement in progression-free survival (PFS) and overall survival (OS) in metastatic melanoma harboring the BRAFV600 mutation and represents the first drug of a class that exerts its anti-proliferative activity through inhibition of a highly specific molecular target. GSK2118436 (dabrafenib), the second BRAF inhibitor, in phase I and II trial obtained similar results to vemurafenib. A phase III trial is now ongoing. Taken together, the early clinical development of vemurafenib and dabrafenib clearly confirms that BRAF inhibitors can halt or reverse disease in patients with melanomas carrying this mutation, improving survival times compared with historically standard treatments (chemotherapy and interleukin-2). The clinical development of other new BRAF inhibitors such as RAF265 and LGX818 is now ongoing. Combination strategies of BRAF inhibitors with ipilimumab, an anti-CTLA-4 antibody, and/or MEK inhibitors or metformin are now under investigation in clinical trials.

Vemurafenib.

PMID: 30069761 | Indexed in PubMed

The activating BRAF mutation V600E and related mutations in this codon are most important for the activation of the RAS/RAF/MEK/ERK mitogen-activated protein kinase (MAPK) signalling pathway in melanoma. BRAF V600E mutations have been detected in ~40% of melanoma patients and BRAF V600K mutations in ~5% of melanoma patients. Activation of the MAPK pathway results in continuous stimulation of cell proliferation and inhibits programmed cell death. Vemurafenib (PLX4032) was developed as a low-molecular-weight molecule for the inhibition of the mutated serine-threonine kinase BRAF, and it selectively binds to the ATP-binding site of BRAF V600E kinase and inhibits its activity. The biochemical affinity of vemurafenib for mutated BRAF translates to potent inhibition of ERK phosphorylation and of cell proliferation exclusively in BRAF-mutant cell lines. In animal model experiments, it was demonstrated that vemurafenib achieved tumour regressions in cells harbouring the BRAF V600E mutation. The clinical trials with vemurafenib in unresectable metastatic melanoma in phases I, II and III for patients harbouring BRAF V600E mutations demonstrated all unexpected high objective response rates ranging between 50 and 80%. Median progression-free survival was prolonged from 2 months with dacarbazine to 7 months with vemurafenib, and median overall survival was, respectively, prolonged from 9 to 14 months. A major problem remains in the development of resistance to vemurafenib treatment after several months in the majority of patients, and multiple resistance mechanisms have already been described. Under vemurafenib treatment, about 25% of patients developed cutaneous squamous cell carcinomas of the keratoacanthoma type with low invasive potential and without the occurrence of metastasis. The overall tolerability of the drug was quite good, and many patients remained on treatment for long times. As other solid tumours like papillary thyroid cancer, colorectal cancer, non-small-cell lung cancer and ovarian cancer likewise harbour BRAF mutation, vemurafenib is also tested in these entities. In future, combinations of vemurafenib with other kinase inhibitors and with immunotherapies will improve its therapeutic potential.

Current Treatment of Melanoma Brain Metastases.

PMID: 37627116 | Indexed in PubMed

Melanoma is a type of skin cancer in which there is a strong correlation between its occurrence and exposure to ultraviolet radiation. Although it is not the most common skin cancer, it has the highest mortality rate of all skin cancers. The prognosis of patients is significantly worsened by melanoma metastasis to the brain, which often occurs in patients with advanced disease. The formation and development of melanoma metastases to the brain involve a very complex process, and their mechanisms are not fully understood. One of the ways for metastatic melanoma cells to survive and develop cancer in the brain environment is the presence of oncogenic BRAF mutation, which occurs in up to 50% of metastatic melanoma cases. Before discovering new methods of treating metastases, the overall survival of patients with this disease was 6 months. Currently, research is being conducted on new drugs using immunotherapy (immune checkpoint inhibitors: anti-PD-1, anti-CTLA-4) and targeted therapy (BRAF and MEK inhibitors) to improve the prognosis of patients. In this article, we summarize the current state of knowledge about the results of treating brain metastases with new systemic therapies.

BRAF Mutation (V600E) Prevalence in Mexican Patients Diagnosed with Melanoma.

PMID: 27194985 | Indexed in PubMed

B-Raf is a serine/threonine protein kinase activating the MAP kinase/ERK-signaling pathway. It has been shown that 50% of melanomas harbor activating BRAF mutations, with over 90% being the V600E mutation. The goal of this research was to determine the prevalence of the BRAF V600E mutation in patients from Central Mexico diagnosed with primary melanoma. Skin biopsies from 47 patients with melanoma were obtained from the dermatology department of the Hospital General 'Dr. Manuel Gea González' in Mexico City. For BRAF mutation determination, after DNA isolation, the gene region where the mutation occurs was amplified by PCR. Subsequently, the presence or absence of the V600E mutation was detected by Sanger sequencing performed at the private molecular diagnostic laboratory Vitagénesis in Monterrey, Mexico. Of the 47 patients sampled, 6.4% harbored the V600E mutation. No statistical significance was found between mutations and the type of tumor.

The role of BRAF V600 mutation in melanoma.

PMID: 22554099 | Indexed in PubMed

BRAF is a serine/threonine protein kinase activating the MAP kinase/ERK-signaling pathway. About 50 % of melanomas harbors activating BRAF mutations (over 90 % V600E). BRAFV600E has been implicated in different mechanisms underlying melanomagenesis, most of which due to the deregulated activation of the downstream MEK/ERK effectors. The first selective inhibitor of mutant BRAF, vemurafenib, after highly encouraging results of the phase I and II trial, was compared to dacarbazine in a phase III trial in treatment-naïve patients (BRIM-3). The study results showed a relative reduction of 63 % in risk of death and 74 % in risk of tumor progression. Considering all trials so far completed, median overall survival reached approximately 16 months for vemurafenib compared to less than 10 months for dacarbazine treatment. Vemurafenib has been extensively tested on melanoma patients expressing the BRAFV600E mutated form; it has been demonstrated to be also effective in inhibiting melanomas carrying the V600K

mutation. In 2011, both FDA and EMA therefore approved vemurafenib for metastatic melanoma carrying BRAFV600 mutations. Some findings suggest that continuation of vemurafenib treatment is potentially beneficial after local therapy in a subset of patients with disease progression (PD). Among who continued vemurafenib >30 days after local therapy of PD lesion(s), a median overall survival was not reached, with a median follow-up of 15.5 months from initiation of BRAF inhibitor therapy. For patients who did not continue treatment, median overall survival from the time of disease progression was 1.4 months. A clinical phase I/II trial is evaluating the safety, tolerability and efficacy of vemurafenib in combination with the CTLA-4 inhibitor mAb ipilimumab. In the BRIM-7 trial vemurafenib is tested in association with GDC-0973, a potent and highly selective inhibitor of MEK1/2. Preliminary data seem to indicate that an additional inhibitor of mutated BRAF, GSK2118436, might be also active on a wider range of BRAF mutations (V600E-K-D-R); actually, treatment with such a compound is under evaluation in a phase III study among stage III-IV melanoma patients positive for BRAF mutations. Overall, BRAF inhibitors were well tolerated; common adverse events are arthralgia, rash, fatigue, alopecia, keratoacanthoma or cutaneous squamous-cell carcinoma, photosensitivity, nausea, and diarrhea, with some variants between different inhibitors.

Age-related next-generation sequencing mutational analysis in 1196 melanomas.

PMID: 36938777 | Indexed in PubMed

Melanoma mutational burden is high and approximately 50% have oncogenic mutations in BRAF. We sought to evaluate age-related mutational differences in melanoma. We analyzed melanoma samples in the Genomics Evidence Neoplasia Information Exchange database. Targetable mutations were identified using the Precision Oncology Knowledge Base (OncoKB). We found 1194 patients with a common set of 30 genes. The top mutated genes in patients <40 years old (y/o) (n = 98) were BRAF (59%), TP53 (31%), NRAS (17%), and PTEN (14%); in 40-59 y/o (n = 354) were BRAF (51%), NRAS (30%), TP53 (26%), and APC (13%); and in ≥60 y/o (n = 742) were BRAF (38%), NRAS (33%), TP53 (26%), and KDR (19%). BRAF mutations were almost mutually exclusive from NRAS mutations in <40 y/o (58/59). Mutational burden increased with age, with means of 2.39, 2.92, and 3.67 mutations per sample in patients <40, 40-59, and ≥60 y/o, respectively (p < 0.0001). There were 10 targetable mutations meeting OncoKB criteria for melanoma: BRAF (level 1), RET (level 1), KIT (level 2), NRAS (level 3A), TP53 (level 3A), and FGFR2, MET, PTEN, PIK3CA, and KRAS (level 4). Mutations in melanoma have age-related differences and demonstrates potential targetable mutations for personalized therapies.